

Q.21 Explain ramjet propulsion principle.

Q.22 Write short notes on unclear rockets.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Derive the steady flow energy equation and explain its application in turbines.

Q.24 Explain the Diesel cycle with PV and TS diagrams and derives its efficiency.

Q.25 Discuss thrust augmentation method in jet propulsion.

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2nd Sem / Aeronautical Engg. Subject : Aircraft Propulsion

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 A system that does not exchange matter or energy with surrounding is:

- a) Closed b) Open
- c) Isolated d) Adiabatic

Q.2 The Zeroth Law of Thermodynamics defines.

- a) Pressure b) Temperature
- c) Enthalpy d) Entropy

Q.3 The relation $C_p - C_v = R$ is Valid for:

- a) Real gases b) Ideal gasses
- c) Liquids d) Solids

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Q.4 In an isothermal process, the change in internal energy is:

- a) Positive b) Negative
- c) Zero d) Infinite

Q.5 The efficiency of Otto cycle depends on :

- a) Compression ratio
- b) Expansion ratio
- c) Stroke length
- d) Fuel type

Q.6 The thrust produced by a jet engine is proportional to:

- a) Mass flow rate $\times$ velocity difference
- b) Pressure difference
- c) Temperature difference
- d) Density difference

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define thermodynamic equilibrium.

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Q.8 What is a quasi-static process?

Q.9 State the First Law of Thermodynamics.

Q.10 Define specific heat at constant pressure.

Q.11 What is a Brayton cycle?

Q.12 Define thrust-specific fuel consumption (TSFC).

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Differentiate between closed, open and isolated systems with examples.

Q.14 Explain intensive and extensive properties with examples.

Q.15 Derive the expression for work done in an isothermal process.

Q.16 Explain the limitation of the First Law of Thermodynamics.

Q.17 Draw PV and TS diagrams of an Otto Cycle.

Q.18 Compare spark ignition and compression ignition engines.

Q.19 Explain turbine efficiency.

Q.20 Derive thrust equation for a jet engine.

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